ATOMIC-LEVEL MATERIALS SIMULATION

Module Code: MES306TC

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Fundamentals of Tight-Binding Method & The Slater-Koster Framework

I. The Core Philosophy of Tight-Binding Method (TBM)

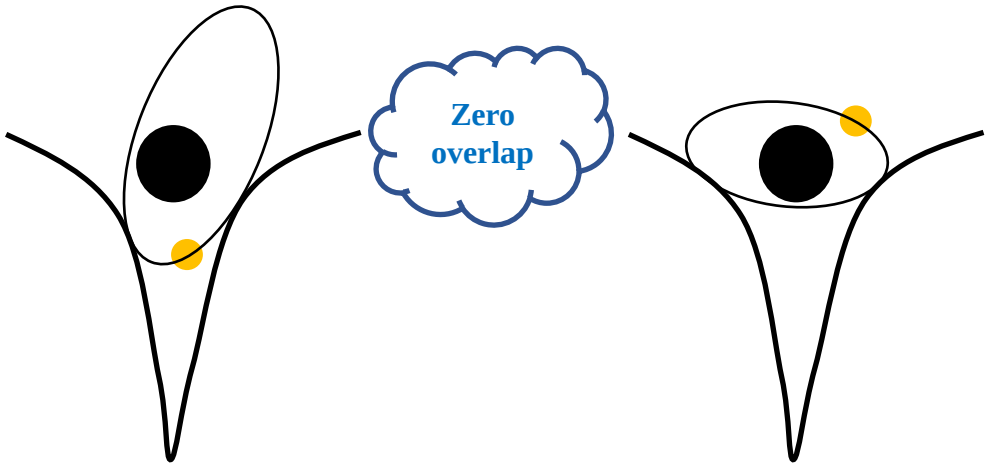
II. Mathematical Formulation: The LCAO Basis Expansion

III. Introduction to the Slater-Koster (SK) Framework

IV. Spatial Truncation: Restricting the Infinite Matrix

I. The Core Philosophy of Tight-Binding Method (TBM)

From Isolated Atoms to Crystals



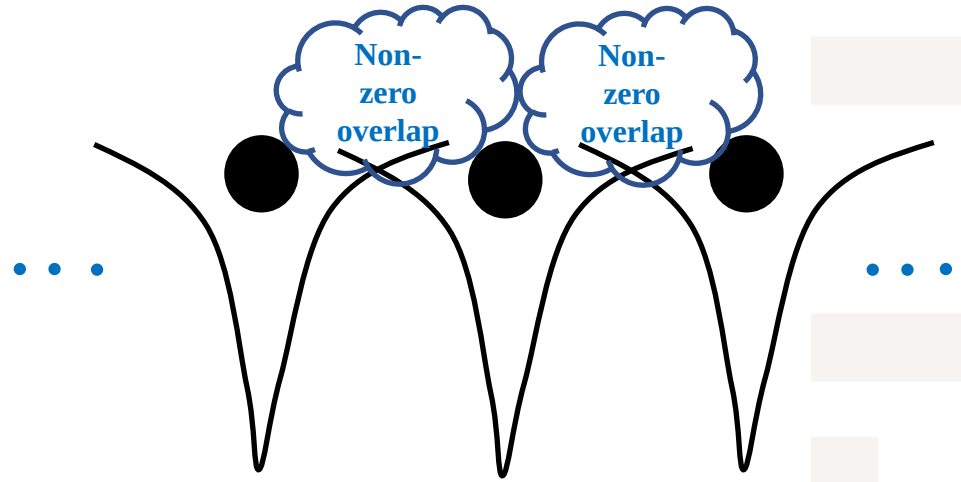
No overlap of potential between each isolated atom

↓

Electrons are trapped within the potential well on each isolated atom

↓

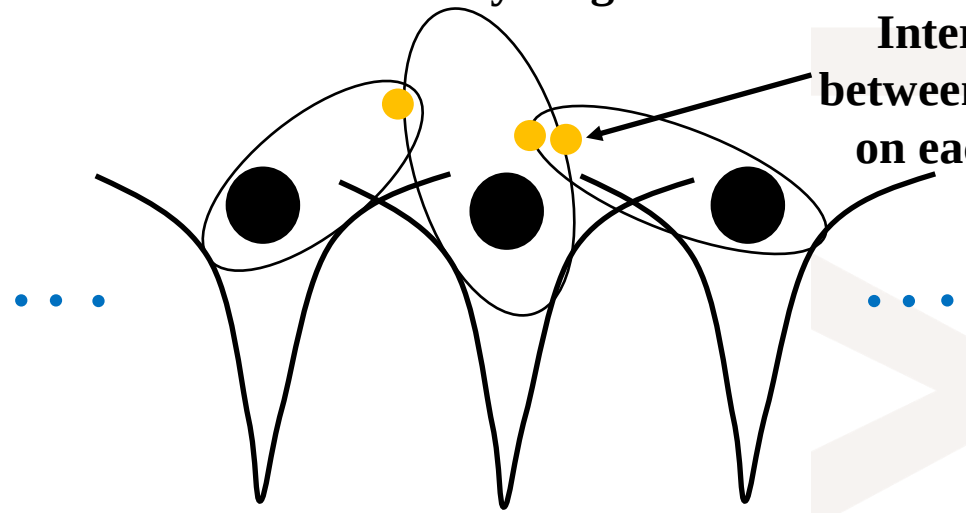
No interaction between electron on each isolated atom



Overlap of potential between each atom

↓

Electrons are not trapped within the potential well any longer



I. The Core Philosophy of Tight-Binding Method (TBM)

From Isolated Atoms to Crystals

Atomic Hamiltonian: $H_{atom}\Phi_{\alpha} = \varepsilon_{\alpha}\Phi_{\alpha}$

↓

No interaction between orbital electrons

↓

Orbitals are discrete and strictly localized

As atoms approach to form a crystal, atomic potentials overlap perturbatively

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$$H_{total} = H_{atom} + \Delta V_r$$

where, $\Delta V(r)$ is the periodic correction potential from all other ions in the lattice

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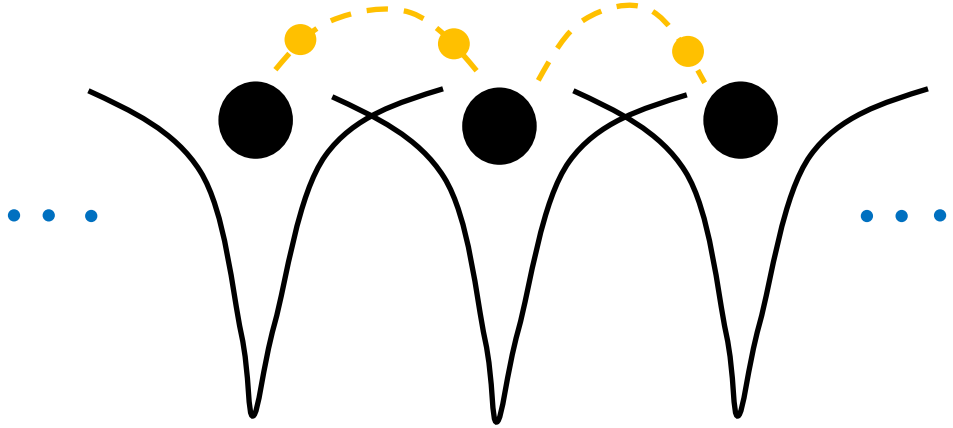
Interaction between orbital electrons

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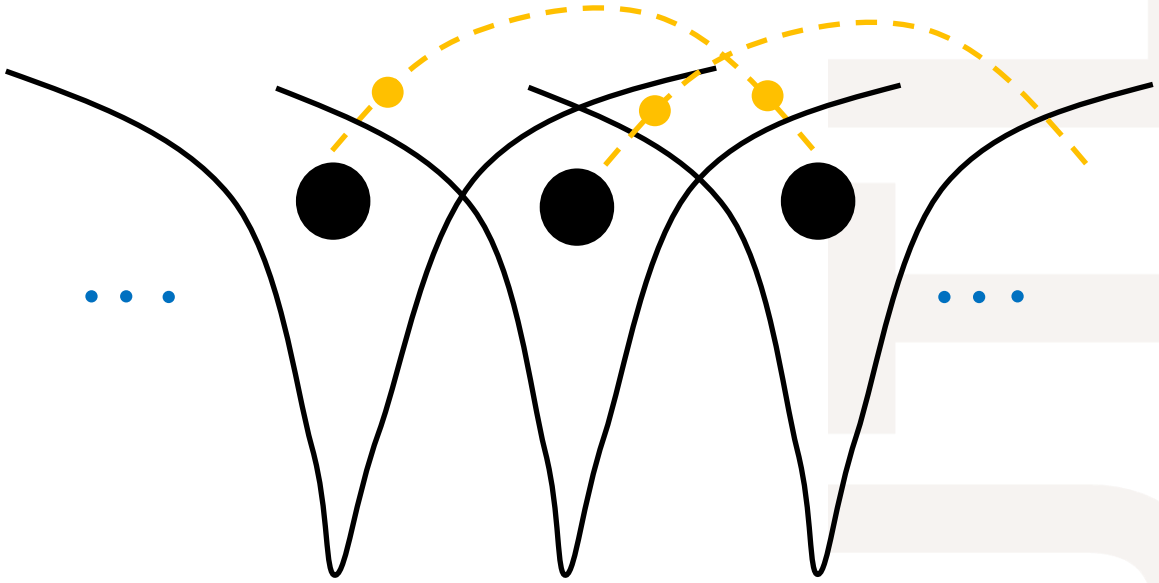
Orbital electrons are not discrete and can jump between atoms

I. The Core Philosophy of Tight-Binding Method (TBM)

Electrons jump between different atoms



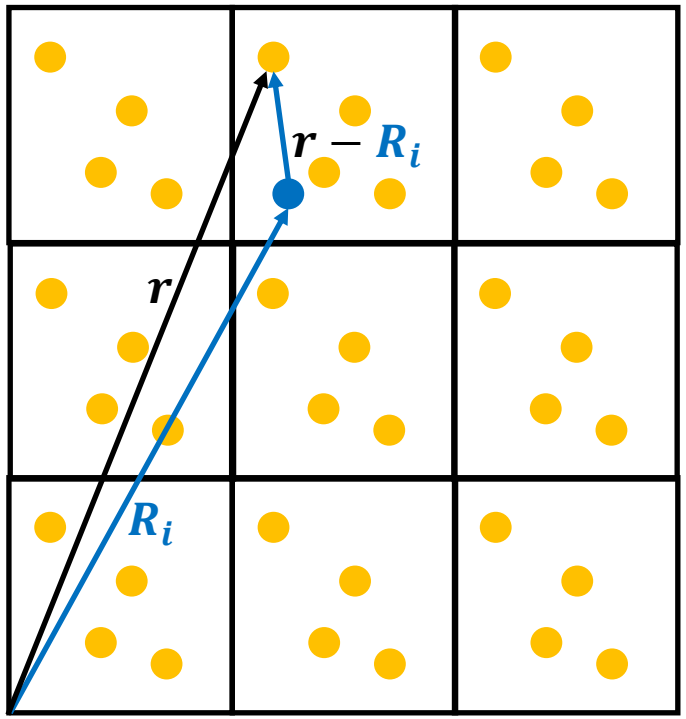
hop between first neighbour atoms



hop between second neighbour atoms

II. Mathematical Formulation: The LCAO Basis Expansion

Crystal wavefunction $\Psi(r)$ is constructed by the linear combination of atomic orbital (LCAO), using a localized atomic orbital basis $\{\Phi_\alpha(r - R_i)\}$:



$$\Psi(r) = \sum_{i,\alpha} c_{i,\alpha}\Phi_\alpha(r - R_i)$$

Matrix Form

$$\Psi(r) = \begin{vmatrix} s & p_x, & \cdots & \\ c_{11} & c_{12} & \cdots & c_{12} \\ c_{21} & c_{22} & \cdots & c_{22} \\ \vdots & \vdots & \cdots & \vdots \\ c_{i\alpha} & c_{i\beta} & \cdots & c_{i\gamma} \end{vmatrix}$$

i **— different position in space**

α — different orbitals (*s*, p_x, p_y, p_z, d_{xy}, …)

II. Mathematical Formulation: The LCAO Basis Expansion

Schrödinger Equation to Matrix Form: $\boldsymbol{H}|\Psi\rangle = \boldsymbol{E}|\Psi\rangle \;\rightarrow\; \sum_{j,\nu} \boldsymbol{c}_{j\nu}\boldsymbol{H}_{i\alpha,j\nu} = \boldsymbol{E}\sum_{j,\nu} \boldsymbol{c}_{j\nu}\boldsymbol{S}_{i\alpha,j\nu}$

Matrix Elements Defined: $\boldsymbol{H}_{i\alpha,j\nu} = \langle \Phi_{\alpha}(\boldsymbol{r}-\boldsymbol{R}_i)\big|\boldsymbol{H}\big|\Phi_{\nu}(\boldsymbol{r}-\boldsymbol{R}_j)\rangle$

Overlap Matrix Defined: $\boldsymbol{S}_{i\alpha,j\nu} = \langle \Phi_{\alpha}(\boldsymbol{r}-\boldsymbol{R}_i)\big|\Phi_{\nu}(\boldsymbol{r}-\boldsymbol{R}_j)\rangle$

II. Mathematical Formulation: The LCAO Basis Expansion

Real-Space Matrix Elements: On-site v.s. Hopping

On-site Energy ($i = j$, $\alpha = \nu$): $H_{i\alpha,i\alpha} = \langle \Phi_\alpha(\boldsymbol{r} - \boldsymbol{R}_i) | \boldsymbol{H} | \Phi_\alpha(\boldsymbol{r} - \boldsymbol{R}_i) \rangle = \varepsilon_\alpha + \Delta\varepsilon_\alpha$

The energy of an electron residing on its host atom, slightly shifted by the crystal environment

$$H_{i\alpha,i\alpha} = \begin{array}{c} \begin{array}{cccc} s & p_x, & \cdots & \end{array} \\ \left| \begin{array}{cccc} c_{11} & 0 & \cdots & 0 \\ 0 & c_{22} & \cdots & 0 \\ \vdots & \vdots & \cdots & \vdots \\ 0 & 0 & \cdots & c_{i\gamma} \end{array} \right| \end{array}$$

II. Mathematical Formulation: The LCAO Basis Expansion

Hopping Integral ($i \neq j$): $t_{\alpha\nu}(R_i - R_j) = H_{i\alpha,j\nu} = \langle \Phi_\alpha(r - R_i) | H | \Phi_\nu(r - R_j) \rangle$

The quantum mechanical probability amplitude of an electron hopping from orbital *v* at site *j* to orbital *α* at site *i*

Keep in mind that the hopping integral *t* usually takes a negative value, which directly stems from the negative sign of our attractive potential ΔV

The very core of TBM: it beautifully converts complex, continuous partial differential equations into a simple problem of solving these discrete matrix elements—namely, the hopping parameters *t* and on-site energies ε

$$H_{i\alpha,j\nu} = \begin{array}{c} \begin{array}{cccc} s & p_x, & \cdots & \end{array} \\ \begin{vmatrix} 0 & c_{12} & \cdots & c_{12} \\ c_{21} & 0 & \cdots & c_{22} \\ \vdots & \vdots & \cdots & \vdots \\ c_{i\alpha} & c_{i\beta} & \cdots & 0 \end{vmatrix} \end{array}$$

$|c_{21}|^2$— Probability of electron jumping from '2' orbital to '1' orbital

III. Introduction to the Slater-Koster (SK) Framework

The Core Conflict in TBM Matrix Assembly

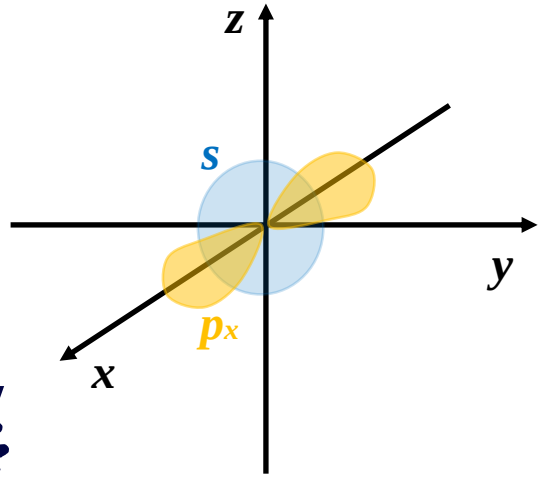
Standard atomic orbitals (p_x, p_y, p_z or d) are strictly locked to the fixed Global Crystal Coordinate System (x, y, z)

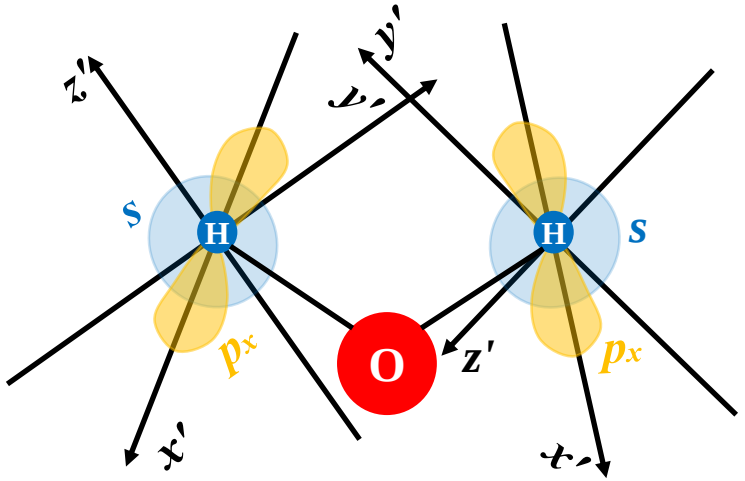


Chemical bonds (inter-atomic displacement vectors) in a real crystal are tilted and point in arbitrary directions throughout 3D space



Calculating direct quantum mechanical integrals $\langle \Phi_{i\alpha} | H | \Phi_{j\nu} \rangle$ for every single unique bond angle and orientation. This makes generalized code implementation computationally impossible





III. Introduction to the Slater-Koster (SK) Framework

The Slater-Koster Solution (1954): A brilliant geometric interpolation technique that decouples quantum physics from spatial orientation

No matter how tilted the bond is, if one looks at the two atoms in isolation, the chemical bonding physics remains fundamentally simple

Axial Symmetry and Local Bond Parameters



The Two-Center Approximation: Assume the hopping interaction depends exclusively on the two host atoms. All three-center potentials (influence from a third neighboring atom) are neglected

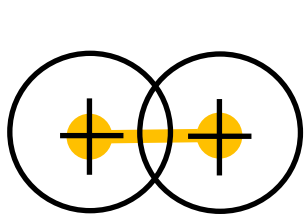


The Local Coordinate System: For any pair of interacting atoms, draw a imaginary straight line connecting their nuclei. Define this bond axis as the local *z*-axis

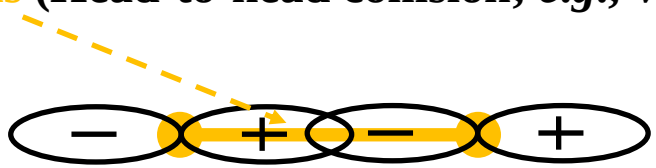
III. Introduction to the Slater-Koster (SK) Framework

Symmetry-Enforced Classification: Along this local axis, cylindrical symmetry ($C_{\infty v}$) dictates that the projection of angular momentum (m_l) is a conserved quantum number. Hopping integrals reduce to pure, isolated chemical bonds

σ bond ($m_l = 0$): Symmetric around the bond axis (Head-to-head collision; *e.g.*, $V_{ss\sigma}$, $V_{pp\sigma}$)

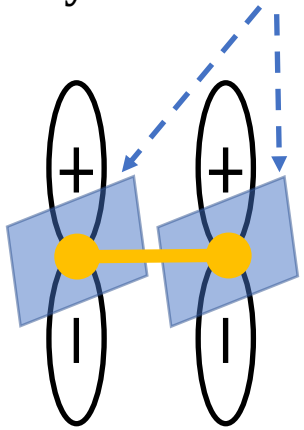


$V_{ss\sigma}$



$V_{pp\sigma}$

π bond ($m_l = \pm 1$): Exactly one nodal plane containing the bond axis (side-by-side overlap; *e.g.*, $V_{pp\pi}$)



$V_{pp\pi}$

Distance Dependence: These basic parameters depend only on the inter-atomic distance $R = |R_i - R_j|$, completely independent of the crystal's tilt angle

III. Introduction to the Slater-Koster (SK) Framework

Step one of the SK method is returning to basic chemistry. If one isolates just two atoms and align the camera exactly along their bond, the physics simplifies due to cylindrical symmetry. There is no messy cross-terms but only the head-to-head smashing, which is called a σ bond, or side-by-side parallel coupling, which is called a π bond. One extract or fit these pure parameters ($V_{pp\sigma}$, $V_{pp\pi}$) as a function of distance. These are the fundamental building blocks.

How to rotate them back into our real, tilted crystal?

Deconstructing Global Orbitals: Tilted global orbitals (p_x, p_y) are mathematically projected onto the local bond axis using rigid spatial rotation matrices

Direction Cosines (l, m, n): The unit vector pointing from Atom i to Atom j in the global frame is given by

$$l = \frac{\Delta x}{R}; m = \frac{\Delta y}{R}; n = \frac{\Delta z}{R}$$

The SK Lookup Formulas: A tilted hopping integral is a geometric blend of pure σ and π interactions, scaled by these direction cosines

III. Introduction to the Slater-Koster (SK) Framework

Example: p_x—p_x Interaction ($E_{x,x}$): $E_{x,x} = \langle p_x|H|p_x\rangle = l^2 V_{pp\sigma} + \left(1 - l^2\right) V_{pp\pi}$

If atoms align perfectly along global *x* axis (l = 1): $E_{x,x} = V_{pp\sigma}$

If atoms align perfectly along global *y* axis (l = 0): $E_{x,x} = V_{pp\pi}$

Step two is pure vector projection. Take the $E_{x,x}$ matrix element as a perfect example.

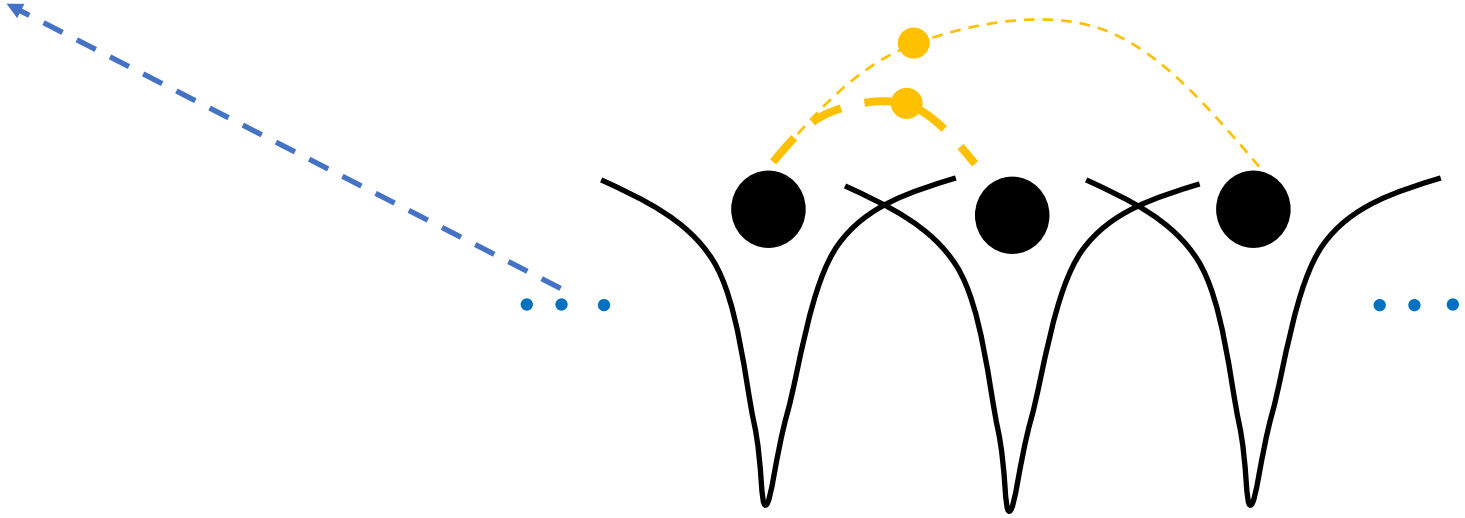
If two carbon atoms are sitting horizontally along the global *x* axis, their p_x orbitals face each other head-on. The direction cosine l is 1. Plug l = 1 into the formula, the π term vanishes and one gets pure $V_{pp\sigma}$.

If they are stacked vertically along the *y* axis, l becomes 0. The formula drops the σ component and gives you pure parallel $V_{pp\pi}$.

What if they are tilted at a 45° angle? Then, l^2 = 0.5. The hopping energy is an exact 50-50 geometric mixture of both σ and π characters

IV. Spatial Truncation: Restricting the Infinite Matrix

Crystal: an infinite-dimensional real-space Hamiltonian



Cutoff Radius (R_c): Since atomic orbitals decay exponentially ($\phi_{(r)} \sim e^{-\alpha r}$), hopping integrals decay exponentially with distance R

$$t_{\alpha\nu}(R) \to 0 \text{ as } |R| > R_c$$

Setting a spatial cutoff enforces Sparsity in the real-space Hamiltonian matrix, making large-scale nanostructure simulations computationally feasible

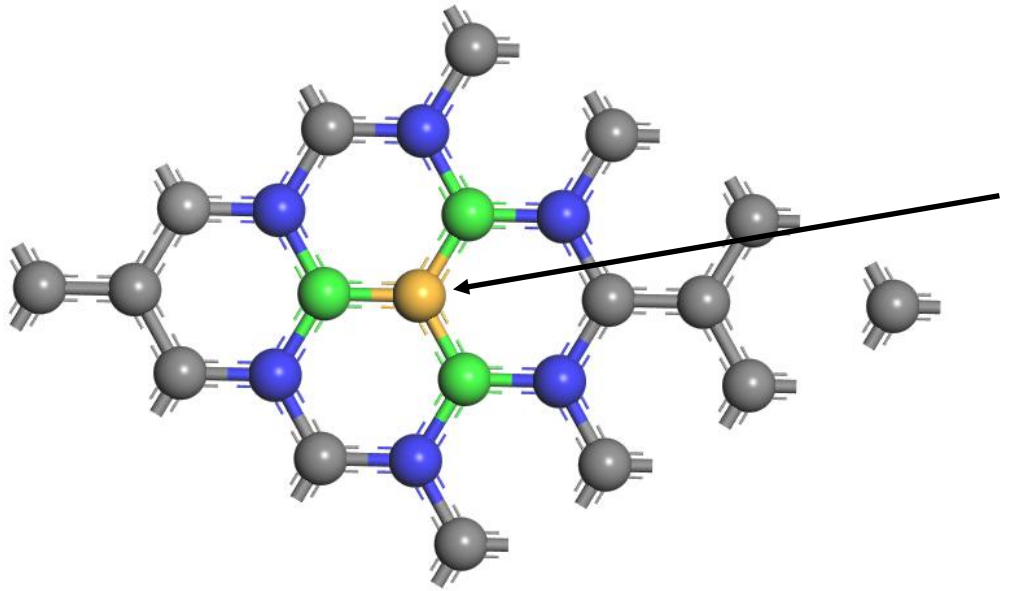
IV. Spatial Truncation: Restricting the Infinite Matrix

The Nearest-Neighbor (*NN*) Approximation

Definition: Only consider hopping between atoms that are immediate geometric neighbors. All $t_{\alpha\nu}$ for 2nd, 3rd and higher coordination shells are set to exactly zero

Mathematical simplification $H_{i\alpha,j\nu} = \begin{cases} \varepsilon_\alpha & \text{if } i = j;\ \alpha = \nu \\ t_{\alpha\nu} & \text{if } i \text{ and } j \text{ are nearest neighbour} \\ 0 & \text{otherwise} \end{cases}$

Orthogonal TBM: On this level of approximation, we frequently assume $S_{i\alpha,j\nu} = \delta_{ij}\delta_{\alpha\nu}$ (ideal orthogonal basis)



Green — 1st neighbour

Blue — 2nd neighbour



That's all for this session

Thank you for your attention and great participation